

Mathematics of Data Science - CA 5

Name: _____

Computation of TV distance

Let P, Q be probability measures on $\Omega = \{1, 2, 3\}$:

$$P = (0.5, 0.25, 0.25), \quad Q = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

1. Compute the total variation distance

$$\text{TV}(P, Q) = \frac{1}{2} \sum_{i=1}^3 |P_i - Q_i|.$$

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4 pts

2. Compute the corresponding ℓ^1 distance

$$\|P - Q\|_1 = \sum_{i=1}^3 |P_i - Q_i|$$

and verify that $\text{TV}(P, Q) = \frac{1}{2} \|P - Q\|_1$.

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3 pts

3. State which metric axioms the TV distance satisfies, and identify one that the KL divergence fails.

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3 pts

Differences:

$$|0.5 - \frac{1}{3}| = \frac{1}{6}, \quad |0.25 - \frac{1}{3}| = \frac{1}{12}, \quad |0.25 - \frac{1}{3}| = \frac{1}{12}.$$

Summing gives $\frac{1}{6} + \frac{1}{12} + \frac{1}{12} = \frac{1}{3}$.

1. $\text{TV}(P, Q) = \frac{1}{2} \cdot \frac{1}{3} = \boxed{\frac{1}{6} \approx 0.1667}$.
2. $\|P - Q\|_1 = \boxed{\frac{1}{3}}$, and indeed $\text{TV}(P, Q) = \frac{1}{2}\|P - Q\|_1$.
3. The TV distance is a *metric* on the probability simplex: it satisfies non-negativity, identity of indiscernibles, symmetry, and the triangle inequality. By contrast, the KL divergence fails symmetry (and triangle inequality), and is therefore not a metric.